

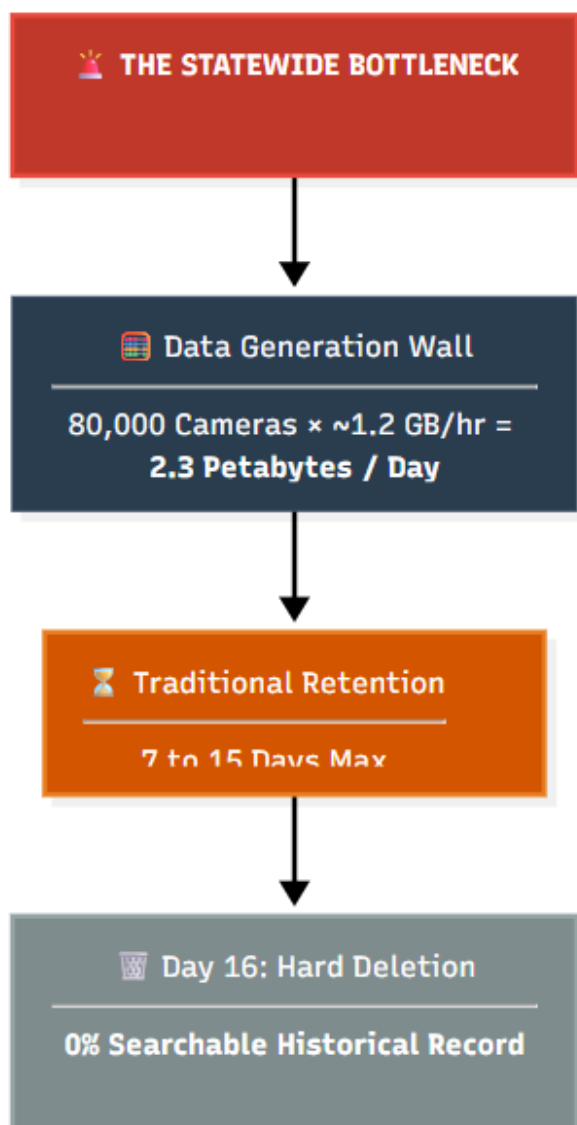
Project Sentinel (BIG EYE) — Statewide CCTV Intelligence Architecture

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"Search the meaning of every camera. Not the footage."

An edge-native intelligence layer engineered to eliminate the statewide storage bottleneck across 80,000 CCTV cameras—making decades of historical traffic, human, and asset movements searchable in milliseconds without retaining a single redundant terabyte of raw video.

1. Executive Summary & Problem Landscape



Modern law enforcement and urban traffic management in Gujarat face a physical and financial limit:

- **The Storage Wall:** Retaining 80,000 high-resolution camera feeds beyond 15 days is economically infeasible.
- **The Investigative Blindspot:** Finding "*a yellow auto-rickshaw carrying suspect cargo last Tuesday*" currently requires human operators to manually scrub hundreds of hours of video per camera.
- **The Disappearance of Evidence:** Once storage policies trigger automated deletion, the historical record of who moved where is permanently lost.

The Sentinel Reframe

Raw video is **99.9% redundant background noise** (static asphalt, walls, and empty roads). Sentinel extracts the **structured semantic lifecycle** of physical objects at the edge the moment they occur and commits a lightweight cryptographic record to a centralized cloud database.

2. Quantitative Impact & Efficiency Proof

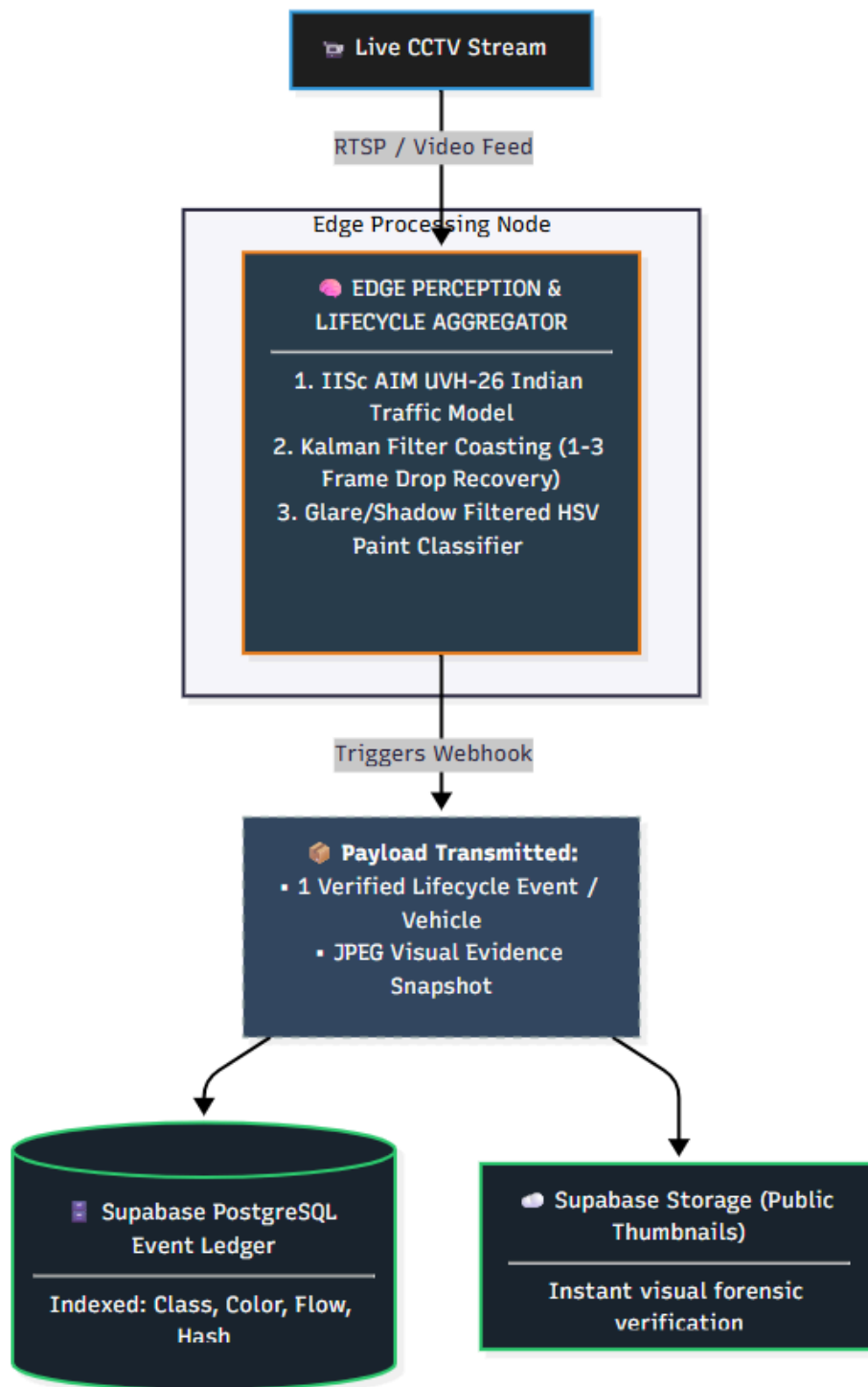
Based on live physical validation on Gujarat CCTV cameras (e.g., `cam01` Chimanbhai Bridge):

Metric	Raw CCTV Footage	Sentinel Metadata Layer	Improvement Factor
1 Hour Footprint (1 Camera)	1,200,000 KB (1.2 GB)	40 KB	30,000× Smaller
80,000 Cameras / 1 Year	840 Petabytes	28 Terabytes	Fits on a standard server rack
Search Time Across State	Hours of manual scrubbing	< 250 milliseconds	Instantaneous SQL / Vector Match
Evidence Retention Window	7–15 days	Indefinite (Years)	Permanent Audit Trail

✓ Business Case

By shifting from *raw pixel retention* to *statewide semantic indexing*, Gujarat Police can retain permanent investigative history across all 26 departments on existing infrastructure at a fraction of cloud storage costs.

3. Core Architectural Innovations



A. The Lifecycle State Machine (Zero-Bloat Ingestion)

Traditional computer vision pipelines write database rows *frame-by-frame* (creating 500 rows for a car visible for 20 seconds). Sentinel introduces an in-memory **Lifecycle State Engine**:

- **State 1: Active Tracking:** Tracks the vehicle across frames, smooths coordinate jitter, and predicts trajectory during temporary frame drops.

- **State 2: Multi-Frame Consensus Voting:** Derives the true class (e.g., Three-wheeler) and true body paint (Yellow/Green), discarding headlight flares.
- **State 3: Event Emission on Exit:** When the object leaves the field of view, the edge node commits **exactly 1 row** containing total dwell time, movement corridor, SHA-256 tamper hash, and an expanded +18% contextual thumbnail.

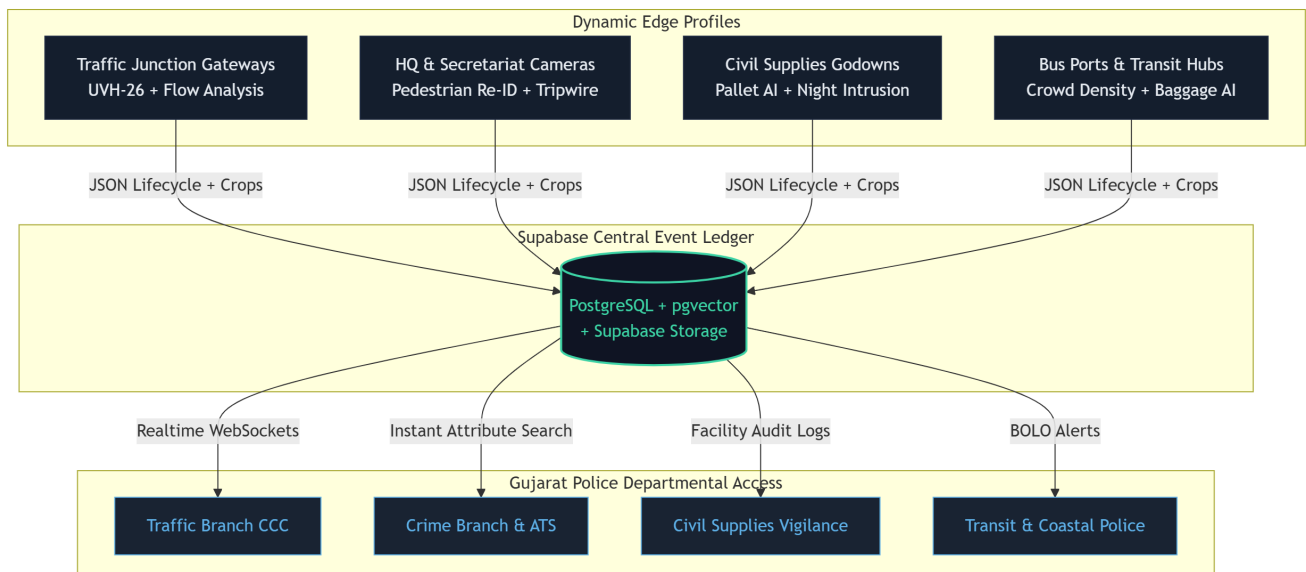
B. India-Specific Perception: IISc AIM UVH-26 Foundation Model

Generic AI models trained on Western datasets (COCO) fail on Indian roads—they confuse auto-rickshaws with small cars, group two-wheelers into blobs, and miss light commercial vehicles (Tata Ace / Chota Hathi). Sentinel directly incorporates the **IISc Bangalore UVH-26 Indian Traffic Model** (31.5% higher mAP on Indian roads), natively supporting 14 fine-grained classes:

1. Three-wheeler (Auto-Rickshaw)
2. Two-wheeler (Motorcycle/Scooter with rider)
3. LCV (Light Commercial Vehicle)
4. Tempo-Traveller
5. Hatchback / Sedan / SUV / MUV / Van
6. Bus / Mini-Bus / Truck / Bicycle / Other

4. Multi-Wing Departmental Matrix (26 Police Wings)

Sentinel supports dynamic edge profiles tailored to the specific operational needs of Gujarat's 26 police departments:



5. Privacy, Ethics & Chain of Custody

1. **Non-Biometric General Surveillance:** Pedestrians are logged only by non-biometric attributes (clothing color, movement vector, dwell time).
2. **Watchlist-Gated Biometrics:** Facial matching vectors are evaluated locally and retained **only upon a direct match** against a sanctioned missing person or suspect watchlist.
3. **Cryptographic Tamper-Evidence:** Every emitted metadata event generates a SHA-256 hash at the edge:

$$\text{Hash} = \text{SHA256}(\text{CameraID} + \text{TrackID} + \text{Class} + \text{Color} + \text{Timestamp})$$

Any post-hoc alteration of database records invalidates the forensic checksum.

4. **Strict Audit Logging:** Every search query executed by an operator is immutably logged with timestamp, badge ID, and query parameters.
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6. Three-Year Phased Roadmap

- **Phase 1 (Immediate / Deployed):**
 - Real-time Indian vehicle & pedestrian metadata ingestion from 30 live Gujarat cameras.
 - Instant attribute search (e.g., "*White SUV inbound on Chimanbhai Bridge*").
 - Storage compression proof ($4,000 \times$ to $30,000 \times$ reduction).
- **Phase 2 (6–12 Months):**
 - Automated cross-camera trajectory stitching (graph-based suspect route reconstruction).
 - Hotspot recurrence analysis (chronic congestion points & loitering alerts).
- **Phase 3 (12–24 Months):**
 - Statewide predictive congestion routing and automated multi-agency BOLO dispatch across 80,000 cameras.